Evaluating Agricultural Loans and Expenditure Allocation in Nigeria Using Double Machine Learning

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Reproducibility Package README

The code within this replication package executes the analysis detailed in the paper "Agricultural Loans and Expenditure Allocation in Nigeria: Evidence from Double Machine Learning" using Stata, R, and Python. This package includes all code required to go from the raw survey microdata to the replication of all tables and figures in the analysis. The Stata master file (`00_master.do`) manages sequential execution of all data construction code. Python estimation is run in a second stage, these programs are computationally taxing and can take between minutes to over a day to run depending on specifications and hardware. Our primary objective is to ensure accessibility and credibility in replicating our research methodology and results.

Data Availability and Provenance Statements

The analysis uses two rounds of Nigeria's General Household Survey – Panel (GHS-Panel), a nationally representative panel survey conducted by the National Bureau of Statistics (NBS) in collaboration with the World Bank. The paper draws on Wave 4 (2018/19) and Wave 5 (2023/24). Raw GHS-Panel microdata are publicly available from the World Bank Microdata Library; free registration is required.

In addition, the analysis uses pre-processed LSMS panel data compiled by the Evans School Policy Analysis and Research Group (EPAR).

Statement about Rights

I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

☑ All data are publicly available.

☐ Some data cannot be made publicly available.

☐ No data can be made publicly available.

Both the World Bank GHS-Panel microdata and the EPAR processed files are freely and publicly accessible at the links above. Place the Wave 4 World Bank files in `Source Data/Nigeria GHS Wave 4/Raw DTA files/` and the Wave 5 files in `Source Data/Nigeria GHS Wave 5/Raw DTA files/`. Place the EPAR processed household files in the corresponding `Source Data/Nigeria GHS Wave 4/epar_data/` and `Source Data/Nigeria GHS Wave 5/epar_data/` folders before running any code.

Data Access Instructions: GHS-Panel microdata, Wave 4:

- **File name:** `NGA_2018_GHSP-W4_v03_M_Stata12.zip`
- **File description:** This zip file contains the Wave 5 General Household Survey-Panel (GHS-Panel) data. Files inside the .zip are typically organized as one file per section within each questionnaire

for either the post-planting or post-harvest period. For example, `sect1_harvestw4.dta` contains responses to Section 1 in the *GENERAL HOUSEHOLD SURVEY-PANEL Wave 4 (2018/19) Post-Harvest Visit Household Questionnaire*. All individual files utilized in the analysis are detailed in Table 4.

- **Citation:** Nigeria National Bureau of Statistics (NBS). 2019. "General Household Survey, Panel (GHS-Panel) 2018–2019, Wave 4." Reference ID NGA_2018_GHSP-W4_v03_M, Version 03. World Bank, Development Data Group [distributor]. https://doi.org/10.48529/1HGW-DQ47. Downloaded from https://microdata.worldbank.org on May 7th, 2026.
- **Access/Download steps:**
  1. Create a free account to access data from the World Bank Microdata Library at https://microdata.worldbank.org/auth/register; after registering and receiving access, login to the account.
  2. On the World Bank Microdata Library (https://microdata.worldbank.org, either use the search-bar to find the applicable repository (code for search-bar `NGA_2018_GHSP-W4_v03_M`) or go directly to the repository via the URL: https://microdata.worldbank.org/index.php/catalog/3557
  3. In the repository navigate to the "Get Microdata" tab (https://microdata.worldbank.org/catalog/3557/get-microdata)
  4. Under the "Get Microdata" tab, navigate to the "Data in Stata format " pull-down, press the "+" sign to see download options, then to download the data click on URL to the right of "Download" (`Download` https://microdata.worldbank.org/catalog/3557/download/47683) or paste the URL into the same web browser you're logged into.
  5. After downloading the data, unzip (extract) the file's contents using your preferred software tool. Once the files are extracted, you can either select the files utilized in our (see Table 4), or the entire unzipped file's contents, to "`Source Data/Nigeria GHS Wave 4/Raw DTA files`" inside the reproduction file.
- **License/access restrictions:** Free access after registering for a World Bank Microdata Library account. Users must agree not to redistribute the files, not to attempt to identify any respondent, and to cite the data.

Data Access Instructions: GHS-Panel microdata, Wave 5:

- **File name:** NGA_2023_GHSP-W5_v02_M_Stata.zip
- **File description:** This zip file contains the Wave 5 General Household Survey-Panel (GHS-Panel) data. Files inside the .zip are typically organized as one file per section within each questionnaire for either the post-planting or post-harvest period. For example, `sect1_harvestw5.dta` contains responses to Section 1 in the *GENERAL HOUSEHOLD SURVEY-PANEL Wave 5 (2023/24) Post-Harvest Visit Household Questionnaire*. All individual files utilized in the analysis are detailed in Table 5.
- **Citation:** Nigeria National Bureau of Statistics (NBS). 2024. "General Household Survey, Panel (GHS-Panel) 2023–2024, Wave 5." Reference ID NGA_2023_GHSP-W5_v02_M, Version 02.

World Bank, Development Data Group [distributor]. https://doi.org/10.48529/90EC-PV69. Downloaded from https://microdata.worldbank.org on May 7th, 2026.

- **Access/Download steps:**

  1. Create a free account to access data from the World Bank Microdata Library at https://microdata.worldbank.org/auth/register; after registering and receiving access, login to the account.
  2. On the World Bank Microdata Library (https://microdata.worldbank.org), either use the search-bar to find the applicable repository (code for search-bar `NGA_2023_GHSP-W5_v02_M`) or go directly to the repository via the URL: https://microdata.worldbank.org/index.php/catalog/6410
  3. In the repository navigate to the "Get Microdata" tab (https://microdata.worldbank.org/catalog/6410/get-microdata)
  4. Under the "Get Microdata" tab, navigate to the "Data in Stata format" pull-down, press the "+" sign to see download options, then to download the data click on URL to the right of "Download" (`Download` https://microdata.worldbank.org/catalog/6410/download/180547) or paste the URL into the same web browser you're logged into.
  5. After downloading the data, unzip (extract) the file's contents using your preferred software tool. Once the files are extracted, you can either select the files utilized in our analysis (see Table 5), or the entire unzipped file's contents, to "`Source Data/Nigeria GHS Wave 5/Raw DTA files`" inside the reproduction file.

- **License/access restrictions:** Free access after registering for a World Bank Microdata Library account. Users must agree not to redistribute the files, not to attempt to identify any respondent, and to cite the data.

Data Access Instructions: EPAR, Nigeria GHS-Panel Wave 4:

- **File name:** `Nigeria GHS W4.zip`
- **File description:** This zip file contains household-, individual-, and plot-level agricultural development indicators constructed by the Evans School Policy Analysis & Research Group (EPAR) from the raw Wave 4 (2018/19) GHS-Panel data. The analysis uses `Nigeria_GHS_W4_household_variables.dta`, which contains household-level indicators.
- **Citation:** Evans School Policy Analysis & Research Group (EPAR). 2025. "LSMS-Data-Dissemination: Nigeria GHS W4." GitHub, commit 0cc96f7 (benchmark code commit 6a431b6). https://github.com/EvansSchoolPolicyAnalysisAndResearch/LSMS-Data-Dissemination/tree/0cc96f7969111a2f4933c53a5d10533c65ce16e8/Nigeria%20GHS. Accessed May 7th, 2026.
- **Access/Download steps:**

  1. No account is required. Click the file name link above, or paste it into a web browser, to download the version of `Nigeria GHS W4.zip` used in this analysis. Note that downloading from the repository's main branch will return a newer version of the data.
  2. After downloading, unzip (extract) the file's contents and place `Nigeria_GHS_W4_household_variables.dta` in "`Source Data/Nigeria GHS Wave 4/epar_data`" inside the reproduction file.

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Data Access Instructions: EPAR, Nigeria GHS-Panel Wave 5:

- **File name:** `Nigeria GHS W5.zip`
- **File description:** This zip file contains household-, individual-, and plot-level agricultural development indicators constructed by the Evans School Policy Analysis & Research Group (EPAR) from the raw Wave 5 (2023/24) GHS-Panel data. The analysis uses `Nigeria_GHS_W5_household_variables.dta`, which contains household-level indicators.
- **Citation:** Evans School Policy Analysis & Research Group (EPAR). 2025. "LSMS-Data-Dissemination: Nigeria GHS W5." GitHub, commit 0cc96f7 (benchmark code commit 6a431b6). https://github.com/EvansSchoolPolicyAnalysisAndResearch/LSMS-Data-Dissemination/tree/0cc96f7969111a2f4933c53a5d10533c65ce16e8/Nigeria%20GHS. Accessed May 7th, 2026.
- **Access/Download steps:**
  1. No account is required. Click the file name link above, or paste it into a web browser, to download the version of `Nigeria GHS W5.zip` used in this analysis. Note that downloading from the repository's main branch will return a newer version of the data.
  2. After downloading, unzip (extract) the file's contents and place `Nigeria_GHS_W5_household_variables.dta` in "`Source Data/Nigeria GHS Wave 5/epar_data`" inside the reproduction file.
- **License/access restrictions:** Publicly available under the BSD 3-Clause License (Copyright (c) 2019, Evans School Policy Analysis & Research Group). Redistribution is permitted provided the copyright notice and license text are retained.

Getting Started

Outline and Folder Structure

The reproducibility package consists of five top-level folders:

- `Source Data/` — GHS survey files (download from the links above).
- `Stata Code/` — All Stata do-files for data cleaning and table production. Final analysis dataset is written to `Stata Code/Stata Data Landing/DML Cleaned Data.dta`.
- `R Code/` — Single script (`FIES Index Calculation.r`) for Rasch-model food insecurity index; called automatically by the Stata master.
- `Python Code/` — DML estimation, sensitivity analysis, and figure scripts.
- `Tables and Figures/` — DML logs, nuisance caches, tables, and final figure outputs.

```
$ROOT/
|-- 00_master.do                  # Stata data and table entry point
|-- 00_figures.py                 # Descriptive and appendix figure runner
```

```
|-- 01_figures.py                 # Main figure runner
|-- requirements.txt             # Python package dependencies
|-- Source Data/
|   |-- Nigeria GHS Wave 4/
|   |   |-- Raw DTA files/
|   |   `-- epar_data/
|   `-- Nigeria GHS Wave 5/
|       |-- Raw DTA files/
|       `-- epar_data/
|-- Stata Code/
|   |-- Stata Data Collection and Cleaning Scripts/
|   `-- Stata Data Landing/DML Cleaned Data.dta
|-- R Code/FIES Index Calculation.r
|-- Python Code/
|   |-- Python DML Scripts/
|   `-- Pthon Figures Scripts/
`-- Tables and Figures/           # Logs, caches, tables, and figures
```

How to Reproduce

PART 1: Stata — Data Construction. Open `00_master.do` from the project root and run it without modification. The script constructs paths automatically from the current working directory. If an error is returned, comments inside should guide troubleshooting, but there are two primary issues that may arise:

1. If `00_master.do` returns an error that a Stata .dta file could not be found then either: 1) the data was not stored properly as outlined in Summary of Availability, or 2) the path macros are not working correctly. In the latter case, `global root "`c(pwd)'"` should set the root to the current reproduction folder. If needed, set it manually following the comments near the top of `00_master.do`.

2. If an error occurs because the Rscript does not run properly or because FIES dataset (`household_probs_long.csv`) cannot be found for Subsection 2.2 of `00_master.do`, then the do file is not correctly executing the Rscript. If Rscript is not on your system PATH, set the executable path manually as described in the comments near the top of `00_master.do`.

This installs required Stata packages, calls the R FIES script, and produces `Stata Code/Stata Data Landing/DML Cleaned Data.dta` and supporting Stata inputs used by the downstream figures. Subsequent Python stages read these outputs. Estimated runtime: ≤ 1 min.

PART 2: Python — Install Dependencies. Install all required packages with a single command (see Section 6.2 for the full `requirements.txt`):

```
pip install -r requirements.txt
```

PART 3: Python — Estimation and Output. Run the following scripts in order. No path changes are needed; each script resolves paths relative to its own file location.

- **PART 3.1 — Descriptive figures and summary statistics.** `00_figures.py` (run from `$ROOT`)

Produces descriptive and appendix figures, including Figures A1–A5, and summary statistics. Outputs are saved to `Tables and Figures/`.

- **PART 3.2 — Standard DML results.** `Python Code/Python DML Scripts/DML Identification.py`
  Set `tipping_point = 0` and select the standard outcomes required for the tables and figures, including fertilizer utilization when reproducing its figure. Paper results use `N_REP = 30` and `k_fold_vector = [5]`. Logs and nuisance predictions are written to `Tables and Figures/` and `Tables and Figures/nuisance_cache/`.
- **PART 3.3 — Balanced tipping-point analysis for Figure 4.** Run `DML Identification.py` again with `tipping_point = 1`, `HTE = True`, both core outcomes selected, `k_fold_vector = [5]`, and `N_REP = 30`. This uses the common two-wave farm/nonfarm sample and writes `Tables and Figures/nuisance_cache/tipping_common/tipping_joint_K5_R30.npz`.
- **PART 3.4 — Omitted Variable Bias Benchmarking.** `Python Code/Python DML Scripts/DML Benchmarking.py`
  Computes omitted-variable bounds by re-estimating with each control dropped in turn. Output CSV files are written to `Tables and Figures/`.
- **PART 3.5 — Main figures.** `01_figures.py` (run from `$ROOT`)
  After the standard and tipping outputs exist, this runs Figures 2–6 and the sensitivity forest plot. Figure 4 reads the balanced tipping output from PART 3.3.

Cache behavior. Nuisance predictions are cached under `Tables and Figures/nuisance_cache/`. Subsequent identical runs load the cache automatically when `re_estimate_nuisance=False`. The tipping calculation does not use cached predictions. Set `re_estimate_nuisance=True` after changing the source data, controls, or learners.

Figure and Table Output Conventions:

Figres and Tables produced in the final manuscript will be automatically be saved to "`Tables and Figures`" located at the reproduction package's root folder. Figures and Tables generally have intuitive file names such `Table_A1_balance_table.tex` or `Figure_4_loan_composition_by_reason.pdf` for for Table A1 and Figure 4 in the manuscript, respectively.

Key exceptions exist, the core DML results, TWFE benchmarks, sensitivity analysis, and quartile regressions are produced by `TWFE Robustness Check.do` and `DML Identification.py`. Both files export model output results to .txt files to "`Tables and Figures`" with the prefix `TWFE_coefficients_*` and `DML Identification_*`, respectively.

The TWFE do file exports **all** results to one unique .txt file (per run), appending for each model, formatted as follows:

```
=========================================================================
RUN:  xxx_date and time of script run_xxx
=========================================================================

=========================================================================
OUTCOME: xxx_income variable_xxx | MODEL: xxx
  N = xxx | Clusters = xxx
=========================================================================
                                   coef   std err  t  P>|t|   adj-R2    within-adj-R2
*************************************************************************
  xxx_treatment_variable(s)_xxx     xxx      xxx     xxx        xxx          xxx
.
.
.
xxx_additional model output below_xxx
```

The TWFE output is then manually written into latex for the final manuscript.

The DML analysis is significantly more computationally taxing (the full DML script for the analysis may take anywhere from hours to days to run — depending on hardware accessibility). Therefore, unlike the TWFE results, the DML analysis script is highly modular so a research may run shorter versions of the analysis by changing features such as dependent variables, repetitions, k-folds, and treatment variables. We provide documentation within `Python Code/Python DML Scripts/DML Identification.py` to help researchers navigate this complexity.

Like the TWFE do file, the DML python script exports results to one unique .txt file (per run), appended for each model, and providing details on modular script settings. The output is formatted as follows:

```
================================================================================
Script: xxx_script name_xxx  |  Version: v1  |  Started: xxx_run time_xxx
Tipping point: x | Data: xxx_data source file_xxx
================================================================================
Treatment: xxx_treatment variable_xxx | K=xxx | N_REP=xxx | Refit=xxx

Loading data from: xxx_data source file_xxx
Loaded xxx observations.

Generating folds: K=xxx, N_REP=xxx...
Folds generated.

Base frame: xxx obs (xxx households)

======================================================================
STAGE 2: Loan nuisance estimation
======================================================================

======================================================================
OUTCOME: xxx_outcome variable_xxx
======================================================================
   -> Imputing xxx variables with missing values, adding indicator flags...
   -> xxx obs (xxx households), xxx dropped households ... xxx_additional
    diagnostics_xxx ...

**********************************************************************
OUTCOME: xxx_outcome variable_xxx | K=xxx | N_REP=xxx | N=xxx
**********************************************************************

>>> DML Estimates:
                              coef   std err   t     P>|t|     2.5 %    97.5 %
xxx_treatment variables_xxx     xxx     xxx    xxx     xxx      xxx      xxx
    Coefficients by repetition: [xxx,...]
    Standard errors by repetition: [xxx,...]

>>> Joint partialled-out regression (main + interaction):
    xxx output if doing the heterogeneous model xxx

>>> GATEs by Farm Size Quartile:
    xxx output if doing the GATE (aka quartile) model xxx

>>> Learner Performance:
    xxx diagnostics statistics xxx
.
.
.
xxx_additional model output below_xxx
```

Similar to the TWFE output, the DML output is then manually written into latex for the final manuscript.

Dataset List

Dataset	Description	Provided
`Source Data/Nigeria GHS Wave 4/Raw DTA files/*.dta`	Raw GHS Wave 4 (2018/19) survey modules. (link above)	No (public)
`Source Data/Nigeria GHS Wave 5/Raw DTA files/*.dta`	Raw GHS Wave 5 (2023/24) survey modules. (link above)	No (public)
`Source Data/Nigeria GHS Wave 4/epar_data/*.dta`	EPAR LSMS processed data Wave 4 (linked above).	No (public)
`Source Data/Nigeria GHS Wave 5/epar_data/*.dta`	EPAR LSMS processed data Wave 5 (linked above).	No (public)

List of Tables and Programs

Table / Figure	Program	Output file
Balance Table	`Stata Code/ Stata Data Collection and Cleaning Scripts/ Stata Output Tables Scripts/ Balance Table.do`	`Tables and Figures/ balance_table.tex`
Main DML Results (ATE, HTE, GATE)	`Python Code/ Python DML Scripts/ DML Identification.py`	`Tables and Figures/ DML Identification_v*_*.txt`
Balanced tipping-point results	`Python Code/ Python DML Scripts/ DML Identification.py` with `tipping_point = 1`	`Tables and Figures/ nuisance_cache/ tipping_common/ tipping_joint_K5_R30.npz`; versioned log under `Tables and Figures/ tipping_common/`
Sensitivity Bounds	`Python Code/ Python DML Scripts/ DML Benchmarking.py`	`Tables and Figures/ benchmark_*.csv`
Summary Statistics	`00_figures.py`	`Tables and Figures/ quartile_summary*.csv`
Figure A1 (FIES KDE)	`Python Code/ Pthon Figures Scripts/ DML Figure A1.py`	`Tables and Figures/ Figure_A1.pdf`

Table / Figure	Program	Output file
Figure A2 (Farm-expenditure KDE by treatment)	`Python Code/ Pthon Figures Scripts/ DML Figure A2.py`	`Tables and Figures/ Figure_A2.pdf`
Figure A3 (Farm-size KDE by treatment)	`Python Code/ Pthon Figures Scripts/ DML Figure A3.py`	`Tables and Figures/ Figure_A3.pdf`
Figure A4 (Farm-size KDE by loan formality)	`Python Code/ Pthon Figures Scripts/ DML Figure A4.py`	`Tables and Figures/ Figure_A4.pdf`
Figure A5 (Farm expenditures by farm size)	`Python Code/ Pthon Figures Scripts/ DML Figure A5.py`	`Tables and Figures/ Figure_farm_expense_ by_size.pdf`
Figure 4 (balanced tipping point)	`Python Code/ Pthon Figures Scripts/ DML Figure 4.py`	`Tables and Figures/ tipping_common/ Figure_tipping_point_ common_sample.pdf`
Main figure suite	`01_figures.py`	Figures 2–6 and sensitivity forest plot
Sensitivity Forest Plot	`Python Code/ Pthon Figures Scripts/ DML Figure Benchmark.py`	`Tables and Figures/ sensitivity_forest_plot_ K5_Rep30.pdf`

Computational Requirements

Software Requirements

Stata

Stata/MP or Stata/SE version 17 or later. The following user-written packages are *automatically installed* by `00_master.do` on the first run:

Package	Source	Purpose
`univar`	SSC	Univariate statistics
`ietoolkit`	SSC	IE data management utilities
`winsor2`	SSC	Winsorizing continuous variables
`reghdfe`	SSC	High-dimensional fixed-effects estimation

R

R version 4.0 or later. Called via shell from `00_master.do` for the FIES index computation. R must be on the system `PATH` or its path must be set in the shell command in `00_master.do`. Required package: `RM.weights` (install via `install.packages("RM.weights")`).

Python

Python 3.11 or later (3.12 used for paper results). Table 3 lists all required packages with the exact versions used.

Table 3: Required Python packages

Package	Version	Purpose
`numpy`	2.0.0	Numerical arrays
`pandas`	2.2.2	Data manipulation; reads `.dta` via `read_stata`
`scikit-learn`	1.7.2	Ensemble learners and cross-validation
`doubleml`	0.11.1	DML Partial Linear Regression
`statsmodels`	0.14.6	Clustered OLS for GATE estimation
`matplotlib`	3.9.0	All figures
`scipy`	1.16.2	Kernel density estimation
`joblib`	1.5.2	Parallel nuisance estimation
`pyarrow`	22.0.0	Parquet I/O for nuisance cache

Installing Python Dependencies

Installation (run once from the repository root):

```
pip install -r requirements.txt
```

`requirements.txt`:

```
numpy==2.0.0
pandas==2.2.2
scikit-learn==1.7.2
doubleml==0.11.1
statsmodels==0.14.6
matplotlib==3.9.0
scipy==1.16.2
joblib==1.5.2
pyarrow==22.0.0
```

To install without pinned versions (use if the above causes conflicts):

```
pip install numpy pandas scikit-learn doubleml statsmodels matplotlib scipy joblib
    pyarrow
```

Key Configuration Parameters

Before running Stage 2, review these settings in `Python Code/Python DML Scripts/DML Identification.py`:

Parameter	Paper value	Notes
`tipping_point`	0 or 1	Use 0 for standard results and 1 for the balanced Figure 4 run
`N_REP`	30	Set to $\leq$5 for a faster verification run
`k_fold_vector`	`[5]`	Cross-fitting folds
`N_WORKERS`	6	Set to available CPU cores
`re_estimate_nuisance`	`False`	`True` forces cache rebuild
`HTE`	`True`	Loan $\times$ farm-size interaction
`HTE_GATE`	`True`	Farm-size quartile effects; disabled automatically when `tipping_point = 1`

`N_REP` and `k_fold_vector` must be set to the same values in `DML Benchmarking.py`. The paper was run on Windows 11 with 6 physical cores and 16 GB RAM.

References

Lars Vilhuber, Connolly, M., Koren, M., Llull, J., & Morrow, P. (2022). "A template README for social science replication packages (v1.1)". *Zenodo*. https://doi.org/10.5281/zenodo.7293838

Evans School Policy Analysis and Research Group (EPAR). (2024). *LSMS Data Dissemination: Nigeria GHS*.

https://github.com/EvansSchoolPolicyAnalysisAndResearch/LSMS-Data-Dissemination

Appendix

Table 4: Nigeria GHS-Panel Wave 4 Data Files

Reading script (path from the repository root), then the data files it loads
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Import Wave 4.do
sect1_harvestw4.dta
sect4b_plantingw4.dta
sect4c1_plantingw4.dta
sect4c2_plantingw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp W4 Food Expenditures.do
sect7a_plantingw4.dta
sect7b_plantingw4.dta
sect10a_harvestw4.dta
sect10b_harvestw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp W4 Nonfood Expenditures.do
sect2_harvestw4.dta
sect8a_plantingw4.dta
sect8b_plantingw4.dta
sect8c_plantingw4.dta
sect9b_harvestw4.dta
sect11_plantingw4.dta
sect11a_harvestw4.dta
sect11b_harvestw4.dta
sect11c_harvestw4.dta
sect11d_harvestw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Food Security.do
sect10c_harvestw4.dta
Script: R Code/FIES Index Calculation.r
sect12_harvestw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Shocks Pull.do
sect15a_harvestw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Land Expenses.do
sect11b1_plantingw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Labor Expenses.do
sect11c1b_plantingw4.dta
secta2b_harvestw4.dta
secta11c2_harvestw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Input Expenses.do
sect11e1_plantingw4.dta
sect11e2_plantingw4.dta
secta11c3_harvestw4.dta
secta11c3q12_harvestw4.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Livestock Expenses.do

Continued on next page

Table 4, continued

sect11i_plantingw4.dta
sect11j_plantingw4.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Extension Treatments.do
sect11l1_plantingw4.dta
secta5b_harvestw4.dta

Table 5: Nigeria GHS-Panel Wave 5 Data Files

Reading script (path from the repository root), then the data files it loads

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Import Wave 5.do
sect1_harvestw5.dta
sect5b_plantingw5.dta
sect5c1_plantingw5.dta
sect5c2_plantingw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp W5 Food Expenditures.do
sect5a_harvestw5.dta
sect5b_harvestw5.dta
sect6a_plantingw5.dta
sect6b_plantingw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp W5 Nonfood Expenditures.do
sect2_harvestw5.dta
sect2c_harvestw5.dta
sect3_plantingw5.dta
sect6a_harvestw5.dta
sect6b_harvestw5.dta
sect6c_harvestw5.dta
sect7a_plantingw5.dta
sect7b_plantingw5.dta
sect7c_plantingw5.dta
sect8b_harvestw5.dta
sect9_harvestw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Food Security.do
sect5c_harvestw5.dta

Script: R Code/FIES Index Calculation.r
sect7_harvestw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Shocks Pull.do
sect12_harvestw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Land Expenses.do
sect11b1_plantingw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Labor Expenses.do
sect11c1b_plantingw5.dta
secta2b_harvestw5.dta
secta11c2_harvestw5.dta

Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Input Expenses.do

Continued on next page

Table 5, continued

sect11e1_plantingw5.dta
sect11e2_plantingw5.dta
secta11c3_harvestw5.dta
secta11c3q12_harvestw5.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Expenditure Calculations Scripts/Exp Livestock Expenses.do
sect11i_plantingw5.dta
sect11j_plantingw5.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Extension Treatments.do
sect11l1_plantingw5.dta
secta5b_harvestw5.dta

Table 6: EPAR-constructed data files, Waves 4 and 5

Reading script (path from the repository root), then the data files it loads
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Import Wave 4.do
Nigeria_GHS_W4_household_variables.dta
Script: Stata Code/Stata Data Collection and Cleaning Scripts/Stata Control Covariate Collection and Cleaning Scripts/Import Wave 5.do
Nigeria_GHS_W5_household_variables.dta